

ActInf GuestStream 124.1 ~ Ty Roachford: "PCT vs. FEP: A Comparison"

GuestStream | Ty Roachford | 1:36:34

Hello, welcome everyone. This is ACTIMF Guest Stream 124.1. It's December 4th, 2025. Interesting. 12.4. Good. And Ty Roachford will be presenting on a recent paper, PCT vs. FEP, a comparison between reorganization theory and Bayesian inference. There will be a presentation, followed by some remarks from our panelists and questions. And I will also look in the live chat for any questions. So thank you, Ty and Warren and Andreas for joining, and to you, Ty, for the presentation. Absolutely. Happy to be here. So this is my presentation on recent paper, Perceptual Control Theory vs. Free Energy Principle, Comparison Between Reorganization Theory and Bayesian Inference. I did write this paper with my collaborator, Warren Menzel, and my PI, Dr. Rodrigo Penna. So getting into it, this is how the paper looks. We publish over in MDPI Foundations. And what we did was a full-stack comparison where we looked at the philosophy, the modeling, and the mathematics around both Perceptual Control Theory and the Free Energy Principle. And we're going to get into sort of what these two pictures, what all these pictures sort of mean in the context of the presentation here. So start off with, I think that there are a lot of scientific explanations that I would call principle-based frameworks, where instead of looking at a bunch of data and then creating theories and models based off of that, it sort of flips the whole entire thing on its head. And we're going from principles, theories, models, really theories and models constrained by principles to data that is being generated by those models. So in other words, we use principles to constrain the space of models that we're actually considering when we go out to model something that is in the world. And in physics, this has already been a thing. This has been a thing for sure in both physics and biology. So here are just some examples of that. Conservation of energy, the maximum entropy principle, the principle of least action, and in biology, having things like autopoiesis, self-organization, homeostasis, and also natural selection. So getting into Perceptual Control Theory as a principle-based framework, we can say that the principle of Perceptual Control Theory is behavior is the control of perception. So rather than looking at organisms as if they're, say, computers or particular functions that take an input and give a specific output, given that input, every single time deterministically, rather you should look at organisms as dedicated to controlling what their input is rather than their output. And so the idea is that more like a thermostat that controls the temperature, you can look at an organism like this, in that when the temperature comes into the thermostat, or even the organism as we see on the bottom here, what's happening is it's going to give a response or a behavior that brings it towards, that allows it to affect the temperature such that it minimizes the difference between its internal set point of 72 in this case, and the outside temperature that it's sensing. So in other words, if it's far too cold, if it's 60 degrees that comes in here, and we have a 72 degree set point, well, we're not responding specifically to the fact that it is 60 degrees, we're responding to the fact that there's a difference between our internal set point of 72 and 60. And there would be a

different response if our internal set point was now at 75 rather than 60. So the output is actually depending on this discrepancy between the sense temperature and the internal reference point that is associated with that temperature. And the same thing goes on in organisms, but a bit more complicated, where we have multiple variables, internal variables that we're regulating, such as hunger and temperature and many other things for many more complicated organisms. And these are all in conflict. So the organism doesn't simply try to do them all at the same time, it sort of prioritizes certain ones over the other in certain contexts, which is what makes organisms so much more difficult to understand. But being that you identify something as a control system, it's much easier to then understand it that way, rather than as a domino effect, where it's just an input leading straight up to an output. And this leads me to this notion that perceptual control theory has, it's called the test for controlled variable, where you identify whether something is indeed a control system, or if it's just a computer domino effect like thing. So on the top of this panel here, figure here, I have this input output system causation paradigm, where you can see a cause or a sensory input coming into some organism, which we denote here as a mechanism, and it caused the observed behavior, what you end up

actually seeing the organism do. And typically people label the sensory input as the cause and the effect as the observed behavior that you see. But when you're looking at a control system, this is not really the case. It's more the case that, like I said, because the cause is actually the discrepancy between the sense temperature and the internal reference variable that's happening inside the organism. And the effect that's produced happens inside the organism as well. But there is external side effects or consequences that you can see as a result of what is going on internally to keep this variable, this input in the way that the organism would like it to be. So what you're actually really seeing here, relabeled in this diagram, is that your sensory input, what you thought was a sensory input or the cause is more like a disturbance to the reference variable that is being set in the control system. And again, the observed behavior being side effects of what it does to bring that reference variable into alignment with the sensed variable. Okay. And this leads to something called the behavioral illusion,

where just if you mistake a control system for an open loop, then you're thinking that you've isolated the cause of

some effect in the organism, but you actually have not. You've been misled.

So getting into the diagram here of how perceptual control theory likes to look at organisms and things, natural control systems, control systems that have been not made by a human, but came up through evolution. Well, they break it into system and environment where we're really focused on the measurable quantities that are in the environment and how they're being disturbed and how that ends up changing the behavior of the system that is embedded in the environment. So what we have here is the input quantity, and this could be something like temperature. And then we also have a disturbance to that temperature. So opening a window would disturb the thermostat's ability to regulate temperature because it would sense that their temperature value has deferred. We have an input function here that

takes the sensed variable outside and converts it into a perceptual signal that then compares with the reference signal P^* and produces an error, which goes into an output function that affects then some other variable, which could be somehow the temperature directly, but really it goes through some sort of physical, some sort of physical chain of events that might be causing it to open a window or turn on a furnace, heat up some coils, something like this, which eventually goes down and affects that input temperature, that input quantity that it's using to create its actual perception of what's going on. And so this is the canonical perceptual control theory diagram that we see in the literature.

An interesting thing about perceptual control theory too, just to hopefully really separate the difference between control system approaches and analytical or computer open loop approaches is this idea versus of analytical versus non analytical solutions. So

the outfielder problem really illustrates this well, where it's how do agents track and catch flying objects? Someone hits a baseball, it goes flying. What does an organism actually do to then go and actually track and catch that object? Well, one way of thinking of it, the analytical or open loop computer perspective is it, something hits the ball and it measures the angle, the initial speed, etc.

So forth takes those variables, plugs them into Newton's equations, like literally it has some sort of Newton's equation form machine, maybe it's made out of biological materials, but that's what it does. It measures those initial conditions, puts them in there, turns it forward until the ball lands and goes, okay, I'm going to go to that particular spot, I'm going to run there. And that's how I'm going to catch the ball. And like, maybe this could work. But there's all these sorts of influences from wind, if you don't get your

initial conditions, your initial conditions, measurements, right, and things can just change over time. So it's not a very good solution, you can imagine, Oh, well, maybe let's just measure it through time over time as the ball is flying. Well, like, yes, but again, it's kind of a little bit, maybe expensive a lot, like now you're computing the same problem over and over and over again, in case something changes, just think, so it's not very efficient. It's not a solution that is

I expect to be in organisms, because it's very expensive. And there could be a simpler, cheaper way. And the simpler, cheaper way is a sort of non analytical solution where instead of having this explicit model, you're doing something more like coupling to the system that you're engaging with in a simplified way so that you can achieve the same solution. So in this case, a non analytical solution would be optical acceleration cancellation, where what happens is the person hits the ball, the ball

goes flying and you see the ball and if it's going faster and faster, according to your point of view, then you know, you need to speed up your running, right. And if it's slowing down, because it'll land behind you. And if it's slowing down, then it's going to land in front of you. So that means you need to slow down. So just by speeding up and slowing down, you can make sure that you're right there keeping a constant velocity. And eventually you'll catch the ball when it reaches the position where you're at.

And then you're at.

It's not about the sorts of many possible states that a thing could actually be in. Rather, it's about how the organization of things allow, say, small bits of information to move a ton of mass in a way that moves some cybernetic or purposive system towards achieving its goals.

whether or not that information can get an organism to achieve the particular task.

Okay, so essentially control information and reorganization theory, I feel like have a really good and interesting match here. And it's sort of answering the question, what are looking to answer the question or describing what informational bits move the system towards its goal most quickly? Well, it would be the informational bits that allow you to control sort of the most energy towards your goal. So if you need to do some particular thing, you want to

you want to be able to do it by inputting the least energy and getting the most output back from that. And this is what control information sort of is talking about here. And this is what I believe reorganization theory has organisms looking for. It's looking for these informational bits that will allow the organism to achieve its goal most quickly.

And then there's also this notion of what informational bits that will allow the system to achieve its goal. And then there's also this notion of what informational bits change the goal that the system pursues. I think this is interesting too, because it could be, oh, you're seeking to go get food, but it's really, really hot. So you have to control your temperature and thus do something else that will help you move towards your goal most efficiently.

Okay. So now summarizing perceptual control theory. For PCT, behavior is the control of perception. Organisms are natural control systems.

PCT is grounded in the physical world through insisting that variables be measurable. Typically in PCT, we're creating equations that are very biophysical, physically grounded. We don't have a bunch of different variables that are not measurable. We want to be able to, even if we have to crack open something and look inside, be able to measure the thing that we think is driving the behavior of the organism.

So we should be able to actually find that internal reference variable when we do some sort of dissection.

And PCT uses TCV to differentiate between open loops and closed loops so that you know what it is that you're studying and how you should actually approach it.

And finally, we have PCT, perceptual control theory aims to increase, I would say, the knowledge of control information, the way that things are organized in such a way to be able to achieve that.

So we need to be able to achieve tasks with sort of minimal amounts of bits.

Cool. So now we can break into going into the free energy principle.

And the philosophy behind that.

Okay. So again, another, I would say, principle-based framework.

And the principle here, to put it like very simply, is if something exists, it must be doing something that allows it to maintain its existence.

And level two, that thing that it does can always be characterized as minimizing variational free energy.

Okay. So there's obviously some sort of physical thing that it is doing.

But the idea is that free energy principle often is very instrumentalist, very a bit, it can be, at least researchers

who use FEP can be a bit loose in the way that they're saying that organisms embody generative models.

And so that makes it sort of definitely different from perceptual control theory in that perceptual control theory is very much about being in the actual physics.

We have to know precisely what are the physical things we can point at and measure that correspond to our equations, to our model of the thing.

Free energy principle, they are allowed, they allow themselves, typically researchers allow themselves to be more loose with it.

And I bring up this paper here by Mel Andrews, where they talk about, they titled their paper, the math is not the territory.

And this is great, because it just essentially says that maybe we should look at free energy principle seriously more as a model structure, rather than specifically a model control where we're saying something about the world beyond our senses.

Maybe we should look at the world beyond our senses, maybe we should look at the math, maybe we should look at the math, and then leave it up to the researcher to come through and point at and say, oh, well, actually, I think that these are the things that correspond to these particular beliefs, like this is its prior here, and this is how the likelihood looks, and this is the mechanisms that are doing that.

But it's not something that I often see in the literature, whereas, again, PCT is centered all around it.

Now, getting into the sort of mechanics of the free energy principle, how do they justify saying that if something exists, then we can use minimizing variational free energy to model it in a sufficient way?

Well, it's the Markov blanket construct, where when you're looking at something, it must be maintaining some internal states or some variables, right, that allow you to identify it as the same thing consistently over time.

And so what that allows you to do is break up the system and its environment into external states, internal states, active states, and sensory states, where external states are the states that belong to the environment.

The sensory states are what are responsible for taking the observations of the organism.

The active states allow the organism to actually interact with the environment.

And the internal states is the brain.

It's sort of driving all of the things there, all the physiological variables that are happening there.

And so what this Markov blanket construct does is that it allows you to say the internal states are independent from the external states, given the active and sensory states, but also vice versa, that the external states are independent from the internal states, given these blanket states.

On the right here, we can see the same thing.

The other states will be its children, the parents of its children, but also its parents.

And so that is how you see this little circular Markov blanket visually going on right here.

And so what this does is it licenses us to use Bayes' theorem, where we can talk about the hidden external state that we don't know, or that the organism, I should say, doesn't know.

And we can cast its behavior, whether it's controlling or doing something else, as doing the minimization of free energy to find out what are the hidden states or the world out there beyond my senses that are actually causing my senses.

If I hear a noise, if I hear a noise, and when I'm in the forest, was it a tiger or was it a tree branch that is snapping?

It allows you to cast organisms as doing this inference in order to survive.

And so I have Bayes' theorem here, and it shows the posterior.

And the posterior is one's new beliefs after they have observed some event, like noise, for example.

So how this works is that you have a prior probability of what you think the world is like beyond your senses.

So if we have this very simple world here where we just hear a noise, and we think that what could have caused it is a tiger or a tree branch snapping,

then we have certain odds associated with either of those.

If we have a balanced situation, maybe we give 50% weight to a tree branch snapping, 50% to a tiger being the cause of the noise.

But we could also, given that we are, I don't know, in the forest, and maybe there's more tigers in this area than particular,

we'll raise the probability and say that there's a 75% chance of a tiger, and there is a 25% of the tree branch actually snapping.

Okay?

And this is something that your likelihood actually does for you in this situation.

You start out with these prior probabilities, and then we're using a likelihood, the actual,

with the probability of the observation given this state to help lead us towards new prior beliefs that will be more in line with what we,

with our model of the world, given the actual observations that we see over time.

And so down here, we have this marginal likelihood, and the marginal likelihood is the likelihood and the prior across all possible states,

so that we're normalizing and getting a proper probability for our new prior that we experience.

Okay, so the problem here is that in Bayes' theorem, we can't use Bayes' theorem directly because this marginal likelihood is often intractable or impossible to compute.

When we do this, we can have finite states, and that typically helps, but if we have an infinite range of possible states that we must consider,

like in the case of temperature, when we're from zero degrees Fahrenheit to 90 degrees Fahrenheit,

well, we have a value for 40.36289 Fahrenheit, et cetera, and so forth, all the real numbers from there, which is intractable.

And so because of that, we end up using variational Bayesian inference.

And indeed, the thought is not that things out there are doing perfect inference,

and so variational Bayesian inference lends itself to being a better, more realistic way to actually model things out there in the world, you could say.

At least it is more practical for us to use to be able to understand things out there in the world.

And so I've rearranged the Bayes' theorem, taking the marginal likelihood that's often intractable and making it our target instead,

and moving the conjectured prior, the posterior over here.

And the idea is that in variational Bayesian inference, we introduce an approximate posterior,

and if you were able to somehow guess what is the correct approximate posterior,

then you would also know what is the marginal likelihood, essentially, yeah?

And so this is a way that helps me get intuition for sort of what variational Bayesian inference is doing.

However, then you may ask, oh, well, how do we pick the correct QS?

Like, how do we know?

Because it could be really any value, and maybe multiple values give us a place to get somewhere.

So what free energy principle does is it helps you find a QFS, an approximate posterior distribution, that minimizes your free energy or your uncertainty with respect to the particular model that you're using to do your observations,

and that is this joint distribution that's constituted by the numerator here in Bayes' theorem.

And to just, like, give some sort of fundamentals, I want to go through the mathematics just of the variational Bayesian inference from the beginning,

but very quickly, just a quick sort of run-through to hopefully be able to translate this, again, more intuitively, all in one spot.

Okay.

So what we're looking at here is just the conception of Shannon surprise or entropy, local entropy, a variable.

So if we had this variable here, probability of X, we could say it's probability of heads or tails, and so that's going to be 50%.

And if it's 50%, then we're over here, and our amount of surprise is quite low, right?

And it just gets lower and lower and lower the higher the probability of the state is.

So if we actually had a coin that was totally rigged such that it gave us heads all the time, and we flipped it and saw heads, we wouldn't be surprised at all.

And if we were to flip a different coin, right, and it had a very, very low probability of heads, then we would be very, very surprised to see heads.

And this is what Shannon surprise does for us, is it allows us to quantify our uncertainty about some probabilistic variable given what we see.

Yeah.

And that is just mathed out here, where I have the local entropy equal to the logarithm of the reciprocal probability, which gives you that really nice curve that we saw before.

But then there is also the wider or global entropy that we more normally use, where instead of just considering the state of heads, we consider both the state of heads and tails.

So it's just a summary of telling you, well, how surprised would I be to see heads, and how surprised would I be to see tails?

How surprised would I be to observe generally the state of this random variable?

Yeah.

Yeah.

It is also looked at as the number of bits necessary to encode the particular distribution that the probability represents.

So it's another very cool and interesting interpretation.

And then getting into, finally, the variational Bayesian inference maths, well, it uses the Kuhlback-Leibler divergence to tell the difference between the two distributions.

Between two distributions.

And so the idea here is that we're going to have our distribution that we actually know.

This is just what comes up in the literature.

So it's really helpful to have this KL divergence here raw, where we're just using single distributions rather than joint distributions, as you saw, hopefully, in the quick flash.

In the next slide here, we actually have the joint distributions that we use in variational free energy.

And so this is just a slide to help people along and see where it comes from, where we're using just basic variables.

And now, when we're looking at this and also seeing how we actually derive this sort of KL-Leibler divergence mathematically.

And so when we're looking at this, we're introducing right here our approximate distribution.

This is our approximate posterior.

And then right here, we actually have our posterior that we really, really want.

And we want to know what is the difference between our approximate posterior and our possible, our new posterior, our new beliefs.

And the idea is that, since, again, it's intractable to actually do regular Bayesian inference, then what we do is we find the minimization of free energy of uncertainty with respect to this particular model.

And that entails us actually blowing this up into a term here that we call free energy, where we have the marginal surprise on the outside because it's sort of the intractable part.

And then since because the KL-Leibler divergence can only be greater than or equal to zero, then you're able to use this to minimize free energy and thus find some sort of distribution that minimizes your free energy with respect to your model to make you more accurate given your model.

And I think that's very important that it's not like the global solution.

It's not truly the most correct that you could be about your observation, about the causes of your observation, but it is the most correct you could be about the causes of your observation given your model, given that you have limited constraints to do full Bayesian inference.

So you do this variational sort of inference here to make this happen.

And I put this here because this is the variational Bayesian inference without your policy, without your actions.

This is just simply observing and it's simpler to mathematically derive.

But then we also have on the bottom here, including your policies, including your actions.

And this is essentially demonstrating how you do both parts, the variational free energy, but then also the expected free energy where variational free energy says, OK, what do I believe is the most likely state of the hidden or of the world, given my observation?

And expected free energy says, well, what do I expect to see given that I do this particular thing?

And given my model?

And given my model?

And this allows the organism to reach its preferred observations, its observations that it would like to see in the world that would help it survive and move through and do all of the things that it does.

OK.

So.

Here I'm just showing that we have the free energy here on the left and we can break it down or look at it another way as the expectation of the difference of logs.

And that we can further say, OK, well, we're taking the expectation with respect to the approximate posterior distribution.

Well, if we take that away, then we have this sort of general setup where we can go, OK, this is how you get free energy.

Well, we'll take the expectation with respect to the approximate posterior distribution of the hidden state given the policy.

And that gives you this sort of thing where you can take the generative model component and break it down into these two probabilities, which give you this interesting interpretation of the model where it's complexity minus accuracy.

So how accurate I am versus how much I am actually changing my beliefs and you're sort of having a balancing act there, which is why they call this an optimal way to update one's beliefs.

The variational Bayesian inference sort of gives you a way to look at that mathematically and make this interpretation.

I feel like in a way that you couldn't do with just Bayes' normal formula.

At least the intuition doesn't seem for me to be there.

We can also take this and do an expectation with respect to the generative model like this.

And we end up getting this equation here, which with some approximations, the approximate posterior from the variational free energy calculation.

And as well as taking instead of, well, our, just our policy and the consequences that it has on our observations, we'll think about, well, what are the preferences that I have and the consequences that this has on our observations?

What is the pragmatic value that I'm getting there? And so this is the information gain.

How much am I learning from this observation?

If I change my beliefs like this versus how much does it actually matter for me in particular?

So there's this, do I go and get the information just because I will know more or will I go and get it because it will help me survive?

And that's the sort of pragmatic value option that's happening here.

And this just brings it all together into a complete slide so that hopefully it can be a helpful resource for someone out there.

Okay. Summarizing free energy principle. So for free energy principle, the behavior of organisms is driven by what any organism expects will allow it to achieve its preferred observations or its goals.

And that this is sufficiently described by Bayes rule or variational inference expected free energy minimization.

And I think the FEP is best looked at as a model structure with no ontological commitments and that the commitments should come from the modeler.

At least you can, you can look at it this way. In other words, you could do something like tests for controlled

variable where you're actually measuring things and identifying whether this is a control system.

And then you're building a generative model about perhaps the beliefs that this organism has about the reference variables it's controlling.

I don't think that you need to use straight up differential equations after that.

And so again, you can use the math and not have ontological commitments saying that this thing out there that I'm modeling really has beliefs, et cetera, and so forth, and is really doing inference in some sense.

Cool. So jumping into the modeling section, we have modeling bacterial chemotaxis.

The idea with modeling bacterial chemotaxis is that it's ubiquitous in biology.

You see it everywhere, not bacterial chemotaxis in particular, but chemotaxis in general, where in the brain we have it controlling where our neurons actually end up during development in the immune system navigating towards viruses and things like that.

Even with inside a cell, you have mitochondria that are following ATP gradients.

And so chemotaxis is everywhere in biology, but it is extraordinarily simple.

It's like well understood, but really, really, really important.

It works by just comparing the difference in concentration of some particular molecule at two different positions in space.

So for the case of the E. coli, for example, you can imagine that we have a sensor right here and a sensor right here.

And due to that, it can know whether it's going in the direction that is increasing the concentration gradient that it's seeking or decreasing the concentration gradient that it's seeking.

So interestingly, bacteria cannot actually choose a particular direction that they're swimming.

They must randomly tumble.

What they do is they trigger all of their phagella.

It causes them to do this sort of spinny thing.

They pick a new direction and they go that direction until error increases.

And they do sort of, you could call it reorganization again to find a better policy or path or thing to be doing such that it allows it to minimize its free energy or eventually reach its goal of no difference between the reference variable and the sensed variable.

And so not only can it not control where it's swimming, but it also experiences this thing called rotational diffusion, where due to the medium of the liquid that it's in its viscosity, it'll be swimming in some direction and it'll get disturbed randomly and that'll shove it off course.

And so it'll have to sort of deal with that too.

And it does. We see E. coli, they do their thing and they call this a bias random walk that it looks really random, the path and trajectory that it's taking, but it gets to its target and it gets to it much more quickly than if it were actually going in some random direction all over the place simply due to rotational diffusion.

And so what we did was make models of this in both perceptual control theory framework and free energy principle framework.

Yeah.

And this is a diagram that illustrates perceptual control theories model.

Dr. Richard Markin and Dr. Bill powers, they wrote a paper on modeling bacterial chemotaxis with PCT, and

we reproduce that reproduce that in lockstep using typescripts, making some programs in the browser.

And this is what it looks like.

So you have X, Y, and these are just your positions of the E. coli.

We have the E. coli in red right here, and we have its actual swimming angle where it's going in this direction.

And we can also calculate the angle that is going towards this target, which is you can think of it as a sugar gradient, and it doesn't quite actually, it doesn't start collecting food until it gets into this area, which just gave, gives us a metric to kind of think about how much, how close it is.

How much time it spends towards the center.

So this gradient here reflects the actual concentration of food, but outside of it, there is no such concentration.

We're only using this angle system to get it towards there.

So in some sense, this is not like a biophysically accurate, realistic model of bacterial chemotaxis.

It's phenomenological.

And we're using this functionality here as sort of a surrogate for how bacteria actually sense using sensors at the front and back of themselves and navigate towards this thing.

Another point here to make just early is that we don't actually have rotational diffusion going on.

So there's no real disturbances that are going on when we have the E. coli bacteria navigating towards its target.

Now we have this position here.

We have the angle that it's actually swimming as α , we have the angle towards the target.

And the perception is this difference between the cosine, between the actual angle and the angle that it's swimming times its velocity times this G constant factor, which in the literature was just one.

So you can pretty much ignore it for the purposes of this here.

And then we have our error, which is simply the comparator where mine, we're subtracting the perceived signal from our reference signal.

We're using that error to accumulate a kind of accumulated historical error.

And so we add this up down here where we're adding DQ to Q .

And if Q is larger than Q_{th} , some capital Q , some threshold, then the bacteria tumbles.

And the idea is that the bacteria needs to tumble in such a pattern or such a frequency that it is able to navigate towards the center of the chemical concentration gradient.

And next thing is modeling bacterial chemotaxis with FEP.

I'm going to show the results of how each of them look later after we go through these details.

And so here we have the likelihood of the generative model that this bacteria embodies.

And we have the prior probability of it being in a certain place.

And so for the states, we just have near to target, mid to target, far from target.

At first, our prior state probability is that, well, we don't know where we are.

So we're going to spread our beliefs, our confidence over all of the states.

And here we have the likelihood, which actually tells us how likely is the observation given this prior knowledge.

And so you'll see that I have the near at the top here, but then also this angle here.

And so what this means is that, OK, well, if I'm facing between 0 and 120 angles to the target, then how close do I believe I am to the target?

Do I believe that I'm far, near, or mid from it?

Similarly here, when I'm looking in this direction, my swimming angle is in this direction or in this direction.

Well, like how exactly, you know, confident am I in my belief that this angle actually represents how close I am to my target?

And then we have this equation down here, low equals one minus high divided by two.

It allows you to make a sort of proper probability matrix where you can have, say, high is 90.

So we're going to have 0.9 and this will be 90 and then this will be five and this will be five.

Similarly, this will be five and this will be five.

So it turns it into a proper probability distribution where you're peaked at a particular point, meaning that you have high confidence.

So I think that the places that high goes because we could sort of rearrange these constitute different hypotheses about sort of where, how likely I believe I am to be close to the target given the angle of my swimming direction in relation to the target.

So, let's see.

So, let's see.

So, let's see.

So, let's see.

And here, this is a precise example of it.

This is a precise example of it right here where we have the actual values filled in.

Cool.

So, we also have what's called a transition matrix.

And so the transition matrix is what's used in our expected free energy calculations where we're wondering about, given that I'm taking this action versus that action, well, what do I expect this state to actually be so that I can come up with what do I expect my observation to be given that particular state.

And so, this is a transition matrix for approach where I have 1 minus NB, NB, NB over 2, 0%.

The idea is that this is a non-locality bias and sort of just like where the placements of high inside of the likelihood matrix give you a sort of different hypothesis about your beliefs of the angles relationship, your swimming angles relationship to how close you are to the target.

Well, this gives you a sort of idea of what's your belief in the world being non-local.

So, if you have a high non-locality bias, right?

And in this case, I've actually set these straight up to 0% so that there is no chance that you have too extreme of a non-locality bias.

So, if you believe you're near on the first time step, then you're not likely to believe that you're, you're, you're 0% likely to believe that you're far on the next time step after taking an approach action.

And so, this is the transition matrix for approaching, and it doesn't mean necessarily approaching the target, but it means for going to continue in the same direction that you are going.

Okay.

Then we have the transition matrix for the tumble action, where after tumbling, we have 33% across all of them.

So essentially, you have no sort of bias towards where you believe you will be in relationship to the target after tumbling.

It's anybody's guess.

And then I have these log preferences here.

And so this is the template log preferences where these preferences are not probability distributions.

They are log prior preferences.

And so only the differences or the relative magnitudes of the values inside of this vector are what count or what matter.

And the more positive, the more favorable, the more negative, the more unfavorable.

So in this situation, since we have 0.00 corresponding to 0 and 20, then we have a preference for going in this direction.

And we don't prefer to go these other directions.

Again, not going away from but towards the actual target.

Cool.

And so there are a lot of different trajectories that came out between these two models.

This is wobbly and then stuck, where we have perceptual control theory, E. coli on the left side, and the POMDP or free energy principle, E. coli on the right side there.

And this is wobbly and then stuck, but we also have wobbly and then loosely stuck.

And we also have very straight and then stuck.

And trajectories like very straight and then loosely stuck.

I took all of these and collected them by hand after simulating a bunch of them and trying to look at, okay, how much do the behaviors of these two models actually overlap?

So that way we can get to a really solid comparison of maybe which one might be better than the other and be able to then choose what explanation we'd like to do going forward.

And I think that there are many trajectories that are unique to PCT or unique to POMDP, but I wasn't able to actually quantify that in a way that is more quantitative rather than qualitative, rather than just kind of handpicking out different things.

And I attempted to use a machine learning algorithm.

I think that that would be really, really helpful.

But this was outside of the scope of what could be done inside the paper with the time that we had deadlines.

So these are the results quantified for PCT, where you can see the distribution of the average distance from the target.

We're only looking at the PCT E. coli that were actually able to reach the target.

And similarly here, we can see the distribution of food collection time ratio, which is essentially how long did it spend inside that blue target, but also how long did it kind of spend towards the center of it?

How stuck was it after it actually got there?

And we have the same thing going on for the FEP results quantified.

And the thing is about these is that I feel that the conclusions are hard to draw.

It's hard to draw any broad conclusions from this due to the fact that I could not do a sort of quantitative estimate of it.

And the thing is that I was able to pick parameters that would make PCT win or FEP win this competition very, very easily.

It seems that any results that I had were super highly dependent on the parameter space.

And for that reason, I feel not comfortable with being able to justify one or the other, given this situation.

Okay.

I think I want to say here that there are there's definitely future opportunities to do better comparison.

We can look at sorts of different metrics and perhaps actually get some sort of quantitative way to, for instance, analyze the trajectories of both PCT and FEP.

And instead of doing it qualitatively like I did, where I just picked through ones that had similar trajectories, you get this principled way of actually binning all of the trajectories by trajectories into different categories that are independent from the model that actually produced it.

That way you could say, oh, these are the parameters that produces these types of trajectories in the PCT model.

And these are the parameters of the FEP model that produce trajectories that are in that same category.

And then when you do that, I think you'd be able to make a more solid, fair comparison.

Because at least you'd know that they are having the same sorts of movements.

Okay.

Okay.

So getting into the mathematics of it, of the mathematical comparison of the two, PCT versus FEP, I chose to use category theory.

Category theory is called like a mathematics of mathematics.

It's super simple and general.

And for that reason, it's just highly expressive.

I'd like to call it sort of, yeah.

Yeah.

So right here, this is what makes up a category, any category in category theory.

And this is just a toy example of a category.

There are serious categories like the category of set or the category of groups.

And here we have objects as different types.

And so types are, it could be, I like to think of it as programming language types where you can have Booleans, integers, et cetera, so forth.

But it can be sort of anything that you're considering to be an object that you have being well defined.

So for example, we could have people here and people have cats.

Yeah.

And we have integers and we have Booleans.

And so this morphism that is a function that essentially tells us what people, how many cats do they have

when I'm given a particular person.

And then you have another morphism here.

G is integer greater than two.

Well, then how many cats do they have?

Maybe it's three.

Is three greater than two?

Right?

It is.

And then so you get a Boolean output as a result.

And the composition of G follows F, these morphisms would be, do they own more than two cats?

And I'd like to call composition like a program.

And associativity is like syntax.

So when you have G follows F, say you could have G follows F follows H, and you could have parentheses around G follows F, but also parentheses around F follows H.

Well, then depending on where you put those parentheses, does actually change the object, whether or not the object is the same.

And so this is what associativity or syntax is doing.

You need to have these sorts of coherence rules to make you have a proper category.

Objects, morphisms, composition, and associativity allows you to do that.

And this is sort of, you could call it the category of cat owners or something like this.

I like that category theory is so expressive that it's quite easy to make sort of diagrams that are rigorous and formal, yet not entirely, completely mathematically rigorous.

And it gives you a way to sort of zoom in to specifically what do you mean by people, the people object in this thing, and make it more complex, more rigorous.

And so in that way, the category theory that I'm about to show you here is not the very mathematically rigorous category theory, but more like how people have pseudocode, where they write up pseudocode before they write up some very specific program.

Well, you can have sort of pseudocategory theory, where you use this to develop real, more intense, rigorous category theory.

And so that's what we're going to get into next, after talking about these couple structures that are helpful tools from category theory, where once you have an actual category, let's call it category ABC, where we can have category XYZ2.

And then we can have a functor between them, where we're mapping the objects and morphisms from one category into the objects and morphisms of another category.

Furthermore, we can have morphisms between functors, and these are called natural transformations, which help you understand the ways in which different categories.

So if functors tell you the relationship between different categories, and you can have multiple relationships between different categories, natural transformations tell you how those relationships themselves are related.

Okay.

So I created this PCT agent category here, where I have the input quantity that is taken by the input function

into the perceptual signal, which is then taken into this sort of tuple here, where we have our reference and our perceptual signal, from which we calculate the error, then pick an action.

And finally, the output function leads us to our output quantity, which affects what I put here, calling it the generative process, not a term that perceptual control theory typically uses.

We call it the environment, but in general, I've made all of these arrows black to indicate that they are part of the environment and not part of the organism that we are considering, so that you can see this sort of environment or organism coupling just here inside of the diagram.

Okay.

And then we also have our disturbance.

Okay.

So the idea being that the environment sort of produces the disturbance that then affects our input quantity that our perceptual control theory organism is controlling.

So now this is the free energy principle agent category, where I am showing variational Bayesian inference in this categorical form.

And so we have the generative process here, which outputs the observation.

And that's why I put the GP in the perceptual control theory diagram to just kind of get out a little bit of similarity to hint more at between, hey, you guys are calling this, this, and they're calling this, that just so we're all aware.

Generates this observation.

Generates this observation.

We observe this perception.

Yeah.

And we create this observation and prior state tuple here, if you will, where we remember our prior belief about what the possible hidden state of the world is.

Then we want to imagine new beliefs about what could have possibly caused this thing, this observation that we see in a way that allows us to be more consistent with our model.

And so we take this S_n , which is Q , which is our approximate posterior distribution, or, yeah.

And we put that there.

Finally, we get into our variational free energy where we're coming up with the hidden state that we believe caused our possible hidden state that we believed caused our observation.

And with that, you can calculate the expected observation, what do you expect to observe given that hidden state.

And then you can consider the expected observation by putting it into this tuple here, or you're going to consider it during your next action.

And so down here on this other path, which is like the goal path, something like this, where you're triggering your memory of a preference.

And we're bringing in the preference here, you're remembering your prior preference, or combining it into this

tuple here, and we get our preferred observation.

What is it that we'd actually prefer to see, not just what is it that we expect to see.

And then we remember this preference.

And together with the preference that we remember, and what we expect to actually see, we calculate our expected free energy, which tells us given this situation,

what can I do, what do I expect will move me towards my preferred observations in the future?

What actions should I possibly take?

And so this gives you a sort of expected free energy, a list of your actions paired with quantities that tell you which one of them is most likely to minimize your,

which one minimizes your expected free energy given your model of the world.

And then you take that action, it influences the generative process, and you continue the loop all over again.

I also have this category here, and this is the category I put of Hodgkin-Huxley neurons.

And so this is just to show that when you're doing perceptual control theory, we are talking about biophysical equations too.

It's not just the diagram there.

Whereas when we're talking about the free energy principle, the diagram that you saw previously is, that's pretty much it.

We're not super often concerned with a mapping between that free energy principle diagram and some particular biophysical equations.

And I think it would actually be really, really hard to do that, but is definitely an interesting endeavor.

And I know that there are, there is work in actually understanding how, for example, neurons encode different likelihoods and priors, etc. and so forth.

And so this diagram shows the category of Hodgkin-Huxley neurons, where we're looking at the neuron as a circuit, and it's controlling its voltage is the best way to look at this.

Neurons have a resting potential, and when they're deviated away from that particular voltage, they always end up coming back to it due to the ionic flows that are happening inside of the neuron and across its membrane.

So we can look at this now, where we see the Hodgkin-Huxley neuron diagram next to the PCT agent category diagram.

And show how we have a functor that maps from the elements inside the Hodgkin-Huxley neuron diagram to the objects that are inside the perceptual control theory diagram.

And this gives you a way to understand Hodgkin-Huxley neurons in terms of perceptual control theory, where you say, okay, I know what is the input quantity, what is the output quantity, things that I could potentially measure, and I understand that what is causing, what is the reference signal that's causing it to actually end up back at that resting potential that I see it characteristically returning to every time it is disturbed by some particular current, from whether that's the outside or internally generated.

current. And this is really what you're dealing with when you're talking about modeling with perceptual control theory, is that you have to have this whole side of biophysical equations or equations that you're using to model the particular parts of the thing that you then relate to this diagram.

Okay. So now we get into the actual comparison between free energy principle and perceptual control theory

here, where I illustrate that with this functor here, it's kind of like a forgetful functor, maybe you could call it where we forget uncertainty.

And the idea is that we're just losing all of the parts where we're explicitly handling uncertainty because that information and perceptual control theory would just be handled by sort of like entirely different models, it'd be oh, which one do I pick and I don't actually write that down and represent it in my model itself.

And you can take this functor and you can take this functor and break it into sub functors which when glued together will give you back this original thing but it helps you clarify different aspects of how these two categories are related so you can see the internal variables in the free energy principle category being mapped to the internal variables in the perceptual control theory category.

And I think there are similar things happening over here for perception and action as well as uncertainty and certainty, where this shows you precisely how we're forgetting the uncertainty handling, in a sense, and this shows you just sort of, I would say maybe they're more similar than different, at least when it comes to perception and action, something like that.

In any case, these reveal different facets to the comparison that has been made by this FEP to PCT functor.

So, a potential synthesis then I see lies in the dual interpretation of the free energy principle where instead of looking at organisms as if they have beliefs about the outside world, well you can flip it and talk about the beliefs that you have about the organisms that are then going and doing things.

And so, I think that this would be sort of like modeling our beliefs about control information about how these things should be particularly organized.

So, instead of beliefs reflecting things that I might not be able to measure, I make beliefs about the equations that I know have variables that I can measure.

And you could sort of think about the beliefs from functors from the biophysical equations to PCT diagrams.

Well, I have the Hodgkin-Huxley diagram and I have functors from it to PCT, but then I could have another different sort of neuron model with functors again to PCT, etc., etc., and so forth.

So, I have a whole set of them.

And then I make some sort of belief probability distribution over my understanding, my mapping of my description of the outside world and sort of my conceptualization of the outside world, too.

So, it's beliefs about the organization of biophysical equations, I think, also is another thing.

So, thinking about perhaps if we have this whole entire pool of biophysical equations and we want to understand something out there that is producing data,

can I somehow make some sort of interesting belief structure in terms of the free energy principle that allows me to take all these biophysical equations and organize them in such a way that I have a model now that is reproducing the data that I actually measured and organized, that I actually measured from the world out there.

And if we could do this, I think it would be really, really cool to be able to generate a bunch of different things that are physically grounded with the biophysical equations using our sort of knowledge that we've codified, but codified inside a particular model structure, something formal, something rigorous.

So, rather than like you could say how maybe PCT and in general folks to do it now, where we have biophysical equations, different ones for every particular thing,

we can have a situation where, and we don't know which one maps to which thing out there,

we can have a situation where we're encoding our beliefs about how we think that these should be organized and be mapped.

And then we're generating models and things, simulated things based off of that.

And then we're seeing the data that they produce and then comparing that to the data that we actually collect in reality.

I think that would be interesting because if you could do this, then you would have a sort of maybe, if not scale free,

at least a way that your models are immediately physically grounded to the thing that you're modeling,

rather than just taking some arbitrary model and fitting it to what you see outside,

because it's not really necessarily going to give you measurable variables, you could do this and you would definitely have measurable variables.

You could go out and actually test your theory as you are generating your theories.

And so future work, elaborating both PCT and FEP models of chemotaxis,

it would be really, really cool to see how hierarchical structures and conflict inside of perceptual control theory map onto the more complicated models of free energy principle using greater depth,

but also hierarchies.

So both wide and tall and deep and seeing how these two things can synthesize in an applicable way.

So developments of hybrid PCT FEP models, but also seeing how both of them compare to each other using category theory.

And hopefully category theory can give a good way for lighting the way for how we can actually synthesize the differential equations,

the biophysical equations that we use in perceptual control theory and physics, traditional physics,

with the sort of, I don't know, new informational physics type free energy principle stuff that we do with the probabilistic generative models.

How can we know exactly how it is that these physical organisms that we see entail or embody generative models?

And how can we ensure that, for example, this physical medium could never have this generative model because of X, Y, and Z,

but this one has, it could only possibly be this generative model because of A, B, and C.

I think that would be really, really interesting and important.

So that's it.

Thank you.

Awesome.

Thank you, Ty.

Thank you.

Thanks, Ty.

Thank you.

I just want to say one more thing.

Being with the Florida Atlantic University, of course, I'm with the Center of Future Mind.

And in Dr. Rodrigo Penna's lab where we study computational neuroscience, co-transmission, biophysical modeling, and things like this.

You can check out my website, Tyfoods for thought, where I have lists of projects that I've sort of done over the years and Eliatic games, which is a way that I started thinking about all this quite some years ago.

It's just a simple toy model where you can think of the biophysical equations and stuff that's going on as being pieces in a game.

And the free energy principle that's organizing all of these pieces is being player in a game.

And so it's kind of an interesting new, very early development of a theory towards making this all cohesive in a simplified manner.

Okay.

Thank you.

Awesome.

Okay.

Warren, first, if you want to provide any remarks.

Thank you.

Thanks, Daniel.

Thanks for hosting this.

And I just want to say, Ty, thank you for such a comprehensive talk.

And it just validates our collaboration over this last year, putting this together.

It's the only person on the planet that can actually articulate these two theories clearly with respect to one another and then communicate it to a wider audience.

So I feel honored for that reason, because there's been so much tension and conflict and misunderstanding on both sides, thinking they're the same, thinking they're completely different, et cetera.

and this piece of work actually starts to identify where those overlaps and those differences are.

I think, I mean, everything that you've said is, everything you've covered is a real foundation to what's going to come next. I just want to clarify a couple of things.

One is that, as we know, the model that you've shown of the PCT unit is just the model of the fundamental unit of PCT,

a control, one control unit. The theory itself proposes that these units are organized in a branching hierarchy of multiple levels

and that there are at least 11 of these levels that Bill Powers has identified as identifying abstracted levels of perception of the outside world.

So, you know, people can look at that in more detail, but I think what it illustrates is that we've compared a subset of PCT, the kind of the basic subsystem PCT with the whole of FEP.

But actually, when you think and you start and you would start to model these units in these complex hierarchies,

and then you have to understand that reorganization, you saw, you had the great diagram there from Bill Powers, showing that the reorganization system actually adjusts the parameters of those systems.

And Bill Powers proposed that what we describe as conscious awareness is the spotlight that finds

those units that it's going to adjust the parameters of in order to update and improve its control. So it opens up to understandings of consciousness, but it also, I think, provides a lens to think about how those parameters, which you had to kind of just cover the whole possible range, but actually within PCT, you've got this reorganizing unit that finds and adjusts the parameters that are going to work for the organism itself. And we haven't really got to that point where we can understand that. And I also think that that once you build that part of the model in, you can see what looks like uncertainty, you can see the model evolving and adapting. And when we see that from the outside, we can see this organism still fumbling around trying to learn, it's having uncertain states because its parameter space is not fully crystallized. And then as that parameter space narrows, it looks like an organism that's getting more certain. But like you say, what it's actually doing is kind of tightening these biophysical parameters to a structure that fulfills its goals and maintains its survival. So I think that's one thing I wanted to say. The other thing I wanted to say is watching your talk now, even though obviously we've worked on the paper together, I was actually thinking of the parallels between you using category theory and what I've kind of had to put into the PCT model, which is a category interface, which was actually suggested by colleague Martin Taylor, a category interface between the perceptual hierarchy and a propositional system, a system that actually codifies the variables in the in the PCT hierarchy as symbols, and then tries to understand the connections and the covariances between all those elements. And I feel that a theory like FEP is much better attuned for understanding those propositional beliefs of a working system than like you're saying the biophysics itself. And so I am starting to see a kind of a fusion, but the fusions for me, it's the PCT stuff's over here in the in the lived hierarchy and the embodied hierarchy. And the the FEP is sitting beside it through a category interface and making these hypotheses about the state of the world in in a more in a kind of more of a reality testing belief testing way rather than a way that's purely about functional adaptation. So yeah, they're my few words, if I hope it makes sense in light of what you've been doing. Absolutely, I would love to know more about that about like, precisely how you're, you know, codifying and building up that system definitely and seeing how Yeah, I'm not doing it with coding or maths, just diet, just pictures and words. But that's always where it starts. Yeah. Very cool. Awesome. Okay, Andres, would you like to give any reflections and questions? Yes, I just want to congratulate you, Ty and the Warren and your PI on a really ambitious project, you know, like, so you really think about the big picture. And I think that you're really providing a service to the world by relating to different worldviews and just seeing, you know, do they connect, can they say the same thing in two different ways? And maybe that says something about just like how different levels of awareness can

interact and so on. I wanted to ask about the category theory. You mentioned the forgetful functor. And I was curious, are you familiar with adjunctions?

Nope, not yet.

So, I've studied category theory, you know, on my own quite a bit. I have a PhD in math.

And I would say, you see, there's kind of like a cult of category theory. So it's, it's a little bit,

I don't know, it's a little bit of funny feeling towards it. A lot of it is verbiage. So it's just

words and words and words and words that really aren't very helpful. But there's a couple of very

deep ideas. So one is the Yoneda Lemma, which is the fundamental theorem. The other is adjunctions.

And so what adjunctions are doing in which you're kind of speaking towards is how do you, it's the

mathematics of analogy. So how do you have a window in one world and a window in another world, and they

match up on that window. Okay. And so there'll be a functor taking you from the one. So instead of

category, I just say world, you know, but you have like a world of free energy principle, you have a

world of perceptual control theory. And so the functor that goes from the free energy principle to the

PCT world is forgetting. It's forgetting, like you said, uncertainty. So now imagine if that same

information exists in both worlds, how do you go back? And that functor is called free construction.

Okay. So there's got to be a way. And it's, it very much relates like to the maximum entropy principle

is saying, look, if you don't know nothing, but you're just going to freely construct things,

right? What would uncertainty look like? You see, what form would it take? What shape would it take?

And then you would automatically get some kind of system that would make sense in the free energy

principle world. You see, so that's a super deep idea. That's a super big win. And because of the

just bold way you're thinking, you get to go there, you see. But when I hear like you talk about pseudo

category theory, et cetera, it's kind of like pseudo Chinese, you know, or pseudo arithmetic,

it's like, you know, it does not, it's like a horse in a tuxedo, you know, it really does not

contribute because when you, a diagram is not a category for many reasons. But the problems is that

the objects you mentioned, you're not clear on what an object is, right? And the morphisms,

they need to be composing. It's not clear what a morphism is, you know, there's huge problems in,

you know, in the world. And it just seems so much more straightforward. If you say, look, we're not

going to mess with category theory, we're just going to work with diagrams. But we're going to

use ideas from category theory, because you don't need to see your strong part was the philosophy,

like your philosophical analysis was very, especially in the paper, just very astute,

you know, and just very informative. This dressing is, it's just so bad. I can't imagine that

why how they could review it and accept it. It's just really bad. So, you know,

you don't need to be hurting yourself that way. One. And so maybe I'll just maybe that's my first

comment. I don't know what, but that's from my heart. That's fair. If you don't think that it,

it's because I called it category theory. Basically, it sounds like it's if I just called it diagrams,

it'd be fine. You said I got diagrams. And this is how I think about my diagrams,

you would save a lot of energy, and you would save a lot of trouble, you wouldn't be stepping,

you know, in puddles. I don't know, you know, why stepping puddles, you know, you just have your own theory of diagrams and your theory diagrams may end up being much more informative than category theory, because category theory has got problems, basically, in terms of how it's developed. So the other thing, maybe to say it's fantastic, like, I think the chemotaxis is interesting thing. But to focus, I think, on how to make it as simple as possible. So the things that you did were absolutely unconvincing and just seemed like not to the point. But I think the idea to say, hey, here's these two different things, how would they model chemotaxis? So for example, one problem is that you're having it go to a point. But what if you just had a, it's really about a gradient, what if you just had a gradient from left to right of the screen, and let's say you have something in the middle, and you send it up. Now, what's the chance it's going to end up going in the right direction in the gradient, right? You see, that should be a much more illustrative problem, because it's just really one, like, can it read a gradient? And then to look at that very philosophically, like, in FEP, you see, it's about memory. If you're going to, it's really a question, like, when they're going through that gradient, are they remembering and measuring that it's weaker now, it's stronger or weaker later, right? Or like with CPT, is it saying, I have a set point, I have a preferred density, right? And I don't care what the gradient is, I'm just looking for my preferred density. That's kind of hard to believe, it seems like they're both trying to want to have, you know, but in order to get the gradient, they have to have a memory, but they would implement the memory differently. And so what's the minimal, you know, problem to say, how are they implementing memory, and then compare that to the actual known biology, you know, how it's about, see, so that would take it, make it real, whereas what you're doing is very, it's like fan fiction, you see, so, you know, is star, I'm going to compare Star Trek and Star World, and how, you know, Star Wars, and how do they compare? Well, a lot of FEP, you know, are you doing fan fiction, or are you doing science? It seemed more like fan fiction. I see what you're saying. That's interesting, because I feel like I addressed both points. I think they're super real. I think that you have to start somewhere. It's like, you draw pictures, and you do words before you get into rigorous mathematical equations and stuff. That's why I said that with pseudocode, pseudocategory theory. I know that Toby McGuire Smith is working on category theory of FEP and like being really rigorous with it, very long epic dissertation that I have not gotten to dive into yet. And then similarly, with the bacterial chemotaxis, it's like, well, I could start with biophysical equations, and really go there that way and get really specific, also with the FEP side, but just starting somewhere and then using that for future work. I'm not saying to use biophysical equations. I'm saying just simple things like just like a little diagram of a bacteria. What do we know? Right? Like, we know that if the things, if the flagella turn one way, they separate and you get tumbling. If they turn the other way, then they unite and you get a, you know, direction in one way. Okay, so there's an on-off switch there, right? What are the receptors like? You see, you don't need to have any biophysics thing, but you need to be able to say, how does memory work? Right? Like, what's the, what's the, what's, how does memory function?

So, um, maybe just to conclude that, um, I think you're doing all these great things, but like maybe the fundamental thing is that with your philosophical analysis, you gave a rather critical view of both. And basically like they both could be wrong or basically not very well-founded. You were both missing the big picture, but then you kind of like, it's just, you're reconciling them, you see, but maybe they don't deserve to be reconciled. You're not really treating them as real scientific theories that could be false, you see. So I think that's the problem is I, I believe they're probably both problematic. Yes, Warren? Yeah, I guess just a couple of things there. I think one, one is that you're, you're right. You're absolutely right. That's the domain, the goal of this project was not to falsify one or other theory. And I think that is a, a, a direction that needs to go where they can be a proper adversarial collaboration where, where both halves, both theorists propose a model in advance of testing the model and that the results of that model, the design is set up so that it can actually differentiate between the two in advance. I think that would be a, a gold standard. The other thing that is a direction for this and Ty and I have talked about this, we were in a kind of a debate about whether to use the Marken and Powers equations or whether to come up with our own. We used the Marken and Powers equations because they were published and associated with PCT. But actually, when you look at how they've, as Ty explained, inferred the perception of the gradient, there's no way that the, bacterium itself through its biophysics could calculate that angle because we've got the privilege of a, of a high level perspective on, on the bacteria. It doesn't. What my hypothesis is that, is that the controlled variable is the, the rate of gradient change. Because if you're going directly towards the, the source of food, then you, the change in, in gradient should be zero. Whereas if you're slightly off track, then as you move, then that gradient should gradually, should gradually dip until when you're perpendicular to the food, for food unit, then it reverses. So, but we haven't been able to, in the timeframe, kind of implement that controlled variable. I think that would be a step towards being a tighter version. But I think if you, if we could then combine that with a, um, kind of more of a pre-registered planned hypothesis test, then we would get closer to this, um, uh, way of testing the two, the two models more directly. The other thing that we've not mentioned is, is just the basics of parsimonie. You know, if two, if two theories can achieve the same ends, which is your preferred theory? For me personally, my preferred theory is a more post parsimonious one. Uh, and we do allude to that in the article as well. And that you really do a good job of the tension between the two theories. And so for me, like, I think that that's a fundamental thing missing by itself on the FEP and maybe on the other theory too, that if you have like two levels of awareness, like unconsciously, a mind that knows the answers unconsciously, but consciously a mind that does not know is divorced from reality. So it's always interrogating, asking questions, right? So you can see those two theories, like one is very physically real and the other one's kind of like very conceptual. And that's a very natural interaction. Um, I would think then the problem is, is that like, which is more relevant, which situation? So yeah, everything could be modeled predictively, but is that really, does that mean that that's how things are right? And I would want to say probably not. So if you were a little bit

more adversarial, like, but sympathetic, but adversarial, you can maybe sort out, like, is there a combination, like you say, they're orthogonal, you know, so how does that fit? And how do they serve each other? But they need to be trimmed down. Both of these theories are probably a little bit more robust than they should be. And they should accommodate the other ones better. That's my thought. Thank you. Thank you.

Thank you.

Yes. Law here, as we sort of come to an end, I think you raised a lot of great points. And I was also very struck by how you took the math is not the territory seriously. And even with Warren's later allusion to that categoric interface, there's this double necessity of cognitive modeling to, on one hand, be the spectator at the baseball stadium watching the outfielder catch the ball, the view from the outside, and then the different analytical ways that that problem could be computed, like as a trajectory with an absolute destination and meeting it. But it's also analytical to reduce the angle of deviation from expectation. So the spectator is always in that analytical mode. And then the spectators are debating, my model is more biophysically grounded because I'm actually modeling this. And so they're having that conversation. But then as you pointed out, like I wondered about the one third breakup or the angle of deviation. And then as you brought up at the end, of course, the bacteria doesn't have access to that angle. So then that's like the low road to active inferences, building joint distributions that cover the whole iguana, the kind of whole cognitive model being its own proof existence that the bacteria does do chemotaxis. And then that openness of modeling from the outside, where if all these different theoretical traditions, different ontologies, sometimes just the same variable with the same name, sometimes same variable, different name, all these pieces that with your wire boarding, you're creating the conditions for there to be a falsifiable discussion around rather than just sort of two ships crossing in the night and giving a starting point for software and approaches to start testing them in terms of utility, but just fitting better or having fewer variables or doing better on a benchmark could never falsify a model in principle. So theory will always be the refuge for those who have a certain abstract perspective. And then on specific settings, there can be the direct comparison of performance of these different models, which kind of brings us to the computer science and AI adjacency of these fields in this very fascinating way, even though they're so deeply grounded in longer philosophical traditions and even embodiment. And so I think you brought together a ton in the paper and

the sort of closing question mode, not to be too committee oriented, but just what are you hoping to see in the coming years of grad school? Like what steps do you want to take moving this forward?

It would be an elaboration of everything that I've done here. So especially the things that Antrius talked about, like going for how exactly, let's get more specific than just the angle towards the thing.

Let's actually look at the biophysics so that even if we don't go fully biophysical, we can talk specifically referring to how this is actually going on in bacteria and then connecting that to generative modeling. And like I said previously, also going more really in depth into the mathematics and the

category theory and like really, really, really fully unpacking the relationship between these two sorts of things and how we can get them to maybe make a hybrid model that's not something cobbled together.

Because right now I sort of imagine it as just like FEP stuff, PC stuff, and really just cobbling it together. And I think that's a, it's a starting point, but it's all really just a starting point.

Um, yeah.

I wanted to, and I like Daniel's comments and I wanted to just add about the baseball thing that, that tension between the subjective view, you were like, you're seeing the ball coming at you and the objective viewer, like you could actually measure the angle if you knew the hole, you know, if you were able to, to, to see it from that angle. See, that's like a common filter. So we're like, you're one person is taking measurements, let's say, you know, by watching the ball, but another person has a theory, they have a theory of, of how balls move, you see. And so you can connect in control theory, you can connect those two types of things. It's very meaningful to have both vantage points.

Hmm. I could see the spectator with their camera or perception, having a common filter, and then also the outfielder with their common filter, and then that categorical interface, and then a sort of total cognitive accounting starts to connect the ontological realism and the biophysical mechanics of what's happening for that bacteria, where there's so many papers on the enzyme phosphorylation and the actual decay rates. But the project and the scope wasn't necessarily to make this atom level biophysical simulation. And that is the question is once you start coarse graining and not just doing the atomistic simulation, you are in the space of the map and the territory.

So, and that's where there's all of these amazing questions and directions for students. So, thank you, Ty. Thank you, Warren. Thank you, Andreas. Thank you.

Good luck with the work and until next time.

Thanks for being part of this. Brilliant. Cheers.

Cheers.

Cheers.

Cheers.

Okay. Be strong. So, thank you. Thank you for this. And I hope to be in touch then somehow, one way or the other. Goodbye. Super. And Ty, we can touch base again.

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